

Gabriel Haw

Portfolio: [ML projects, code, and technical writeups](#)
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PROFILE

Applied Machine Learning Engineer with experience building predictive models on high-dimensional biological and time series datasets. Skilled in feature engineering, model evaluation, and hyperparameter tuning using Python, scikit-learn, LightGBM, and XGBoost. Experienced working with datasets containing tens of thousands of features and millions of observations. Strong focus on reproducible workflows, rigorous evaluation, and translating complex data into quantitative models.

MACHINE LEARNING PROJECTS

Epigenetic Age Prediction from DNA Methylation

Trained and evaluated machine learning models on ~12,000 samples with ~37,000 CpG features to predict biological age. Implemented Elastic Net, LightGBM, and neural network models. Performed feature engineering, dimensionality reduction, and hyperparameter tuning. Evaluated performance using cross-validation and regression metrics. Identified predictive biological signals from high-dimensional genomic data.

Electricity Consumption Forecasting

Built forecasting models on 2M+ observations using Random Forest and XGBoost. Engineered temporal features including lag variables, rolling statistics, and calendar features. Detected and resolved data leakage that inflated model performance. Reduced prediction error by 42.7% compared to baseline. Evaluated models using MAE, RMSE, and realistic validation strategies.

Brain Connectivity Modeling (Master's Thesis)

Analyzed structural connectivity across 77 subjects and ~450 brain regions using diffusion MRI data. Modeled connectivity decay using variogram analysis and extracted quantitative features describing connectivity structure. Identified structural patterns associated with cortical hierarchy and spatial organization.

EXPERIENCE

EcoSolutions, Part-time Data Analyst
ecosolutions.co.za

Johannesburg, South Africa
Apr 2021 – Jul 2023

Analyzed ecological datasets using Python and R to model bat-box occupancy patterns. Performed feature engineering and regression modeling to identify predictive environmental variables. Built statistical models and evaluated relationships between predictors and occupancy rates. Communicated findings through data visualization and technical reporting.

TECHNICAL SKILLS

Programming: Python, SQL, R

Machine Learning: Regression, classification, feature engineering, cross-validation, hyperparameter tuning, model evaluation

Libraries: scikit-learn, LightGBM, XGBoost, pandas, numpy, matplotlib, seaborn

Data Analysis: Time series analysis, statistical modeling, exploratory data analysis, hypothesis testing

Tools: Git, Jupyter, Linux, Tableau

EDUCATION

University of Amsterdam

Amsterdam, Netherlands

MSc Bioinformatics and Systems Biology

2023 – 2025

Grade: 8.4 / 10

Relevant Coursework: Scientific Machine Learning, Deep Learning, Algorithms in Sequence Analysis, Statistics with R, Image Processing and Quantitative Data Analysis

University of Bologna

Bologna, Italy

BSc Genomics

2019 – 2023

Grade: 101 / 110

Relevant Coursework: Programming, Data Science, Statistical Methods for Genomics, Algorithms and Data Structures in Biology, Mathematics